William Pettifer

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Technical Skills

Languages: C++, C#, Python, GDScript, GLSL **Systems and libraries:** OpenGL, OpenTK, SFML, OpenMP, Unity, Godot **Tooling:** Git, Jira, Confluence, Linux, Nsight Graphics, Valgrind/Massif, perf, Blender, Aseprite **Areas:** data structures and algorithms, concurrency, memory management, spatial partitioning, profiling and optimisation, linear algebra, real-time rendering, procedural geometry, adversarial search

Education

University of Warwick | MSc Computer Science · 2026-2027 (ongoing)

University of Portsmouth | BSc (Hons) Computer Games Technology · 2023-2026
First Class Honours, 3.91 GPA. Founded the university Debate Society.

King Edward VI Camp Hill School for Boys · 2014-2022
A Levels: Politics, Psychology, Art (BBB). GCSEs: six grade 8s including Mathematics, plus English and four further subjects at 7-6.

Experience & Projects

Grey Cat Games Ltd| Co-founder and Business Secretary · 2025-present

- Own the full visual pipeline solo: modelling in Blender, albedo texturing in Aseprite for a low-poly pixel-art look, then implementation and shader work in Unity.
- Coordinate timelines and task tracking across the team using Jira and Confluence, all members working alongside other full-time commitments.

Space Dog Fight | 3D combat simulation · C#, OpenGL/OpenTK · 2025

- Wrote every mathematical primitive by hand with no external libraries: vectors, column-major matrices, 3x3 inversion by cofactor expansion, quaternions, SLERP, and a sine approximation using Bhāskara I's formula.
- Simulated stable orbits and binary systems using gravitational velocity and barycentre calculations.

Ant Colony Simulation · C++, SFML, OpenMP · 2025

- Profiled with Nsight Graphics, Massif and perf to locate the true bottleneck before optimising rather than working from assumption.
- Reduced marker removal from O(n) to O(1) after benchmarking three container strategies, using stored grid indices with swap-and-pop; parallelised agent updates with OpenMP.

Pine Engine | PS1-style renderer · C++, OpenGL · 2026

- Designed a recursive node-based scene tree; diagnosed a camera bug caused by the update path treating transforms as local while the render path resolved them globally, and fixed it by walking the hierarchy on demand rather than duplicating state.
- Reproduced affine texture mapping, vertex snapping, colour quantisation and dithering, extended with a multi-light Gouraud pipeline computed in the vertex shader.

Caravan | strategy game with adversarial AI · GDScript, Godot · 2025-2026

- Replaced an array board representation, whose removal operation shifted memory on every call, with a compact string encoding and two-digit integer moves, making apply and undo cheap enough for deep recursive search.
- Implemented minimax at configurable depth and evaluated four agents over 1,200 automated games, confirming the expected hierarchy by depth and measuring a greater than 20x cost increase per additional level. Released on itch.io.

Pipeworks |Godot editor tool · GDScript, Godot · 2026

- Spline-based pipe placement tool running live in the editor via Godot's @tool workflow, with curve parameters exposed through @export for artist-facing tuning.
- Diagnosed mesh twisting as a degenerate cross product where the tangent aligned with the fixed reference vector; replaced the global reference with parallel transport frames propagated along the spline.